Predicting Perceived Intelligence from Design, Behavior and User Judgments

A reproducible secondary analysis of four human studies published in 2026

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Abstract

Voice assistants can feel competent without providing reliable task evidence, and successful interactions can receive poor subjective evaluations. Existing research already models these perceptions; the practical question is what information a predictive model actually needs. We compare predictions based on experimental conditions, user attributes, recorded behavior and concurrent subjective judgments, keeping these information sources separate. We analyze 553 released eligible observers from two voice-assistant vignette experiments, 148 users of a text co-creation system, and 240 participants in a text-LLM latency experiment. These are 941 participant records across four studies, not one harmonized sample. Intelligence and credibility are strongly associated in the voice studies (HTMT 0.847), while two-factor models retain poor absolute fit. Study-held-out design-plus-user prediction explains only 2.2% of intelligence variance in the forward transfer; contemporaneous credibility explains 51.8%. In co-creation, nested ridge and nonlinear models using available interaction logs give little improvement in point estimates over the training mean, whereas same-survey trust predicts about half the variance. In the text-latency data, event counts weakly predict thoughtfulness and usefulness, while other concurrent ratings give R^2 near 0.50–0.55. Manipulation-check filtering changes the pooled voice role contrast from −0.055 to +0.410 points under a first-check rule, demonstrating estimand sensitivity. The contribution is a new reproducible analysis of measurement overlap, information availability and transfer limits, not a new intelligence scale or universally validated voice metric. Live voice behavior, objective task correctness and direct intelligence ratings are not jointly observed; prospective validation remains necessary.

Keywords: perceived intelligence; voice user interfaces; human–AI interaction; measurement validity; cross-study prediction; secondary analysis; reproducibility.

1. Introduction

A useful evaluation model must distinguish what an assistant did from what a person inferred about it. Factual correctness, task completion, response timing, clarification and repair are observable aspects of an interaction. Perceived intelligence is an attribution made by a person. Credibility, usefulness, satisfaction and willingness to rely on the assistant are related outcomes, but substituting one for another changes the research question. This distinction matters when teams optimize a score: a predictor that needs a second questionnaire may reconstruct the first questionnaire very accurately while offering little guidance about which system behavior to change.

The broad idea of modeling subjective quality from interaction features is established. PARADISE linked task success and dialogue costs to satisfaction, SASSI identified multiple speech-interface dimensions, and contemporary signaling accounts already connect assistant

features, attributed intelligence and evaluations. [1] [2] [3] Our aim is therefore deliberately narrower than proposing the first model of assistant “smartness.” We ask how much predictive evidence comes from different information sources, whether neighboring subjective constructs are empirically separable, and how conclusions change under sample selection and external testing.

We contribute three linked empirical results. First, we compare direct intelligence measures with credibility and trust, retaining poor fit and instrument discrepancies rather than declaring independent dimensions. Second, we compare participant- and study-held-out prediction under explicit information constraints, including a nonlinear sensitivity model. Third, we reproduce the source voice study's selected-sample descriptive results and quantify the change associated with filtering on post-treatment role recognition. All analysis uses real, publicly released human observations; no agent-generated dialogue, LLM judge or synthetic participant supplies an empirical outcome.

The scope is consequential. The voice data are evaluations of recorded interactions, not live spoken sessions. The other datasets contain actual human–AI interaction but use text. We do not pool incompatible outcomes, transfer their coefficients to a voice product, or represent the resulting manuscript as a completed prospective voice experiment. The present results concern secondary analysis; conclusions about live interaction require new data.

2. Related work

2.1 Measurement precedes model fitting

SASSI, SUISQ and Godspeed demonstrate why an overall impression should not automatically be treated as one interchangeable quality variable. Godspeed directly measures perceived intelligence alongside other attributions; SUISQ concerns speech-service quality. Their measures answer different questions and need contextual validation. [2] [4] [5] [6] More recent work includes perceived-intelligence models for voice shopping and a travel-assistant scale with conversational intelligence, information quality and anthropomorphism. [7] [8]

The Partner Modelling Questionnaire measures competence/dependability, human-likeness and flexibility. Its validation and cross-cultural adaptations are particularly relevant to “understands me,” but a factor that performs adequately in one setting need not transfer unchanged across languages. [9] [10] Consequently, our use of different item sets in different datasets supplies complementary evidence, not measurement invariance or a newly validated Russian questionnaire.

A targeted follow-up found Moussawi and Koufaris's full 2019 scale-development paper, including an independent survey validation sample. Its intelligence items already span speed, understanding, communication, processing and useful answers. Independent validation and multidimensional operationalization therefore cannot be claimed as our inventions; its survey nomological analysis differs from our held-out information-source contrasts. [11] Zhao and colleagues' primary abstract reports a four-dimensional intelligence model using SEM and fsQCA, and Krämer and colleagues report a six-wave longitudinal study of trust and satisfaction. Their detailed full methods remained inaccessible in our follow-up, so they remain explicit limits on any priority claim. [12] [13]

2.2 Behavior, context and perceived capability

Guha and colleagues already treat observable features as signals of intelligence and artificiality, with user moderators. Xiao and Zhang's 2026 study links control variety and complexity to intelligence judgments across Alexa skills. [3] [14] Neither a larger path diagram nor new names for these mechanisms would establish novelty. Our information-source comparisons operationalize a different question: what prediction remains possible when other subjective evaluations are unavailable?

Contemporary experiments also discourage universal cue rules. Tan and colleagues study delay and task type in text interactions; Maslych and colleagues study fillers during delayed embodied dialogue; Jacka and colleagues examine speech disfluency and partner models. The studies differ in setting, intervention and endpoint. [15] [16] [17] We therefore do not encode “faster always feels smarter” or “fluent always means competent” as fixed assumptions. A 2026 older-adult preprint and recent in-home qualitative work further motivate distinguishing personality, capability, trust and task context. [18] [19]

2.3 Current automatic evaluation

USR and FED establish multidimensional dialogue evaluation; SPUR and PRAISE develop interpretable satisfaction prediction. These are substantive comparators, but satisfaction labels from text dialogue are not direct intelligence measurements. [20] [21] [22] [23] We do not compare our R^2 against published scores on incompatible benchmark tasks or claim state-of-the-art performance.

Novikova and colleagues already predict direct robot-intelligence ratings from linguistic, task and multimodal features in live spoken interaction, using random forests and a held-out split. This is a closer predecessor than satisfaction-only prediction and prevents any claim of the first interaction-feature predictor of intelligence. Their embodied setting and available features differ from our studies. [24]

Ma and colleagues' CHI 2026 work already predicts voice-interface UX from speech features in a 49-person study. Its jointly manipulated personas and evaluation protocol delimit causal and generalization claims. [25] Recent primary industry reports, including DIAL and VOICE-H, provide larger human evaluations of voice interaction or speech output, but their aggregate dashboards and endpoints do not supply a validated individual intelligence target for our analysis. [26] [27] TRACE/SCOPE also separates success from perceived experience, but its synthesized conversations cannot substitute for human observations here. [28]

Our literature search used primary publisher pages, ACL Anthology, arXiv, author and university repositories, and exact-title/backward searches; the expanded prospective review has a cutoff of 5 November 2026. The companion review records access status, queries and unresolved leads. This is a targeted scoping review, not an exhaustive database search with a PRISMA denominator. This study contributes a comparison of measurement and prediction across the selected datasets. The search does not establish that no similar analysis exists.

3. Measurement and prediction framework

Let D denote assigned design conditions, B observed interaction behavior, U reported user characteristics, C scenario or study context, I latent perceived intelligence, and T neighboring subjective evaluations. A future interaction model would use D, B and U to predict I while allowing context-specific coefficients. A separate measurement layer links I to ordinal questionnaire responses. Correlations between I and T are estimated rather than constrained to zero. This is a model specification built from established measurement and prediction methods; its conceptual components are not claimed as new theory.

Measurement: $P(Y_{ij} \le k \mid I_i, T_i) = \text{logistic}(\tau_{jk} - \lambda_j I_i - \gamma_j T_i)$.

Structural specification: $I_i = \alpha_C + \beta_{D,C} D_i + \beta_{B,C} B_i + \beta_{U,C} U_i + b_{participant} + \varepsilon_i$.

The full ordinal hierarchical specification is not identifiable from all available releases: there are no repeated scenario-specific intelligence outcomes in the voice table and no common instrument across modalities. The implemented estimable model uses the mean of explicitly defined intelligence items as its target, a ridge regression for prediction, continuous-item covariance factor models as measurement diagnostics, and a single-item ordered-logit sensitivity analysis in the voice data. The prospective protocol describes how to estimate the richer model from new live sessions. These levels of completion are kept separate.

Three information sets define the empirical comparison. The design/user set contains assigned conditions and reported attributes. The interaction set adds only available logs, such as counts and session duration; it does not invent accuracy, response latency or semantic quality. The diagnostic survey set includes other contemporaneous ratings. The latter is a benchmark for concurrent association, not a deployable upper bound or causal mediator estimate. End-of-session adoption is separately marked because it may follow, rather than cause, a competence judgment.

Ridge: minimize $\Sigma_i (Y_i - \alpha - X_i \beta)^2 + \lambda ||\beta||^2$, with λ chosen inside training data only.

We define the predictive gap as the difference between held-out performance using concurrent ratings and performance using design/user or behavioral evidence. Its interpretation is descriptive: it measures dependence of predictive performance on the available information. It does not identify common-method bias, a psychological mechanism, or an unattainable limit on prediction from richer logs. A latent halo, genuine correlated perceptions, omitted behavior and measurement noise can all contribute.

Five exploratory questions guide the analysis: (E1) how separable are intelligence and adjacent judgments; (E2) does design/user prediction transfer between voice experiments; (E3) how sensitive is the role contrast to manipulation-check selection; (E4) do observed co-creation logs add predictive value; and (E5) do latency and task effects depend on the subjective outcome? The analysis specification was written after source inspection, not externally preregistered. Reproduction checks are documented in research/verification.md; the nonlinear model is an exploratory sensitivity analysis.

4. Data, provenance and estimands

We selected public datasets accompanying 2026 publications with human ratings, interpretable design or behavioral variables, and verifiable primary sources. Selection was opportunistic and access-constrained, not a representative sampling frame of assistants. The analysis date is not the data-collection date. Older commercial voice stimuli and a particular text model remain part of these recent publications.

Voice vignettes. Sarigül, Schneider and Utz provide two experiments of role framing and, in the first experiment, speech style, with UK native-English-speaking Prolific participants. We use 353 and 200 released eligible respondents. Five intelligence and five credibility items are rated on seven-category scales. Six scenarios are reflected in a final evaluation, not six independently labeled rows. Earlier technical exclusions from the second experiment are unavailable. The full-release estimate therefore is not strict intention-to-treat over all recruited participants. [29]

Text co-creation. Komura and Yamada release questionnaires, session metrics and note-level records for 148 Japanese Yahoo Crowdsourcing participants interacting with GPT-4.1 on a brainstorming canvas. The source recruited in May 2025. Adapted five-category intelligence items and 0–7 trust items are available. Six assignments are recorded as unknown: measurement uses all 148, primary condition-aware prediction uses 142, and an all-148 prediction sensitivity is retained. Adoption is a recorded choice, not independently verified usefulness or correctness. [30]

Text latency. Tan and colleagues release 2,651 events from 240 US Prolific participants using a custom GPT-4o interface, assigned to 2/9/20-second delays and Creation/Advice tasks. Seven quality ratings occur once per participant. Three IDs have conflicting task labels among their event rows. Primary inference uses the survey-bearing row; a sensitivity excludes those IDs. Prediction uses the unambiguous 237. These outcomes contain no direct intelligence item. The source already excluded 63 participants using a post-assignment latency-fidelity threshold; their records are absent. These contrasts concern the released sample, not a recovered full intention-to-treat effect. [15]

Table 1. Real human data used. Counts are records across studies; no common-scale pooling is performed.

Source	Released units	Analyzed endpoint	Validation boundary
Voice experiment 1	353 observers	Intelligence, credibility (1–7)	Recorded voice vignettes
Voice experiment 2	200 observers	Intelligence, credibility (1–7)	Held-out related experiment
Text co-creation	148 users; 142 known assignments	Adapted intelligence (1–5)	Participant and condition holdouts
Text latency	240 users; 237 unambiguous	Thoughtfulness, usefulness (1–7)	Participant CV; separate endpoints

Raw files remain unchanged and are identified by SHA256 and source URLs. The SPSS-to-CSV export preserves numeric values; variable labels remain available in the original SAV files. Intelligence composites in the voice release exactly equal their five-item means. In co-creation, all selected-note totals reconcile across the detail and summary sheets. The file has 5,849 notes, one more than the published text, and anonymized note IDs collide; the unique key is session × creator × turn. Each session has exactly 20 AI notes, so that count is uninformative and excluded from final predictors.

Instrument differences are material. The co-creation scale includes responsiveness and omits the usual sensible item; its supplementary likeability block also repeats an adjective pair. The voice preregistration and final instrument differ in one adjective label. We follow the final released instruments and do not harmonize scores through simple rescaling. The original studies' ethics and recruitment apply to their observations. No new human-study approval or consent collection is claimed for this secondary work. PLOS materials and the latency repository carry CC BY terms; the voice repository exposes no explicit blanket license, so its raw data are not relicensed by this project.

5. Analysis and verification

5.1 Measurement diagnostics

We calculate internal consistency, composite associations, HTMT with 2,000 participant bootstrap draws, and one-factor versus correlated-factor covariance models. HTMT and fit indices are diagnostics, not universal pass/fail proofs of separability. Modern methodological work warns against automatic discriminant-validity conclusions from a single cutoff. [31] [32] Covariance models treat ordinal items as approximately continuous; no robust ordinal CFA, scalar invariance or longitudinal invariance was completed. We retain model misfit and do not add residual correlations after inspecting modification indices. Information criteria are calculated relative to the same saturated-model constant and compared only within sample and item set.

5.2 Prediction under information constraints

Voice models train on experiment 1 and test on experiment 2, then reverse direction using the assertive arm of experiment 1 to match the style supported in experiment 2. Testing the reverse model on all experiment-1 participants is a shifted-style sensitivity, not a third independent replication. User features are age as 2024 minus reported birth year, source gender indicators, smart-speaker ownership and mean reported information-search frequency across five named services. Ridge penalties are selected by five-fold training-only CV minimizing MAE. Imputation and scaling are inside the pipeline. Nonowners have structurally missing service-frequency responses; these remain missing until training-only median imputation with missingness indicators.

A subsequent voice sensitivity resamples training and test participants independently 1,000 times per transfer and fully refits both compared models, including penalty selection. Bootstrap copies of the same original participant are assigned to one inner-validation group using GroupKFold. This avoids duplicate leakage, although it changes the original shuffled-KFold tuning protocol. We therefore also fit a grouped-CV reference on the original samples. Percentile intervals incorporate empirical training and test variability; they do not account for new stimuli, study choice or unknown source exclusions.

Co-creation uses five participant-disjoint folds and leave-one-known-condition-out testing. Inner four-fold CV chooses ridge penalties from 0.1, 1, 10 and 100 by squared error. A shallow histogram gradient-boosting sensitivity uses 100 iterations, 3 or 7 leaves, minimum leaf size 15, L2 values 1 or 10, and no validation-based early stopping. All preprocessing and tuning remain inside training folds. Design and demographics, session logs, final adoption, and same-survey trust are compared in predefined blocks. The model does not use the free-text comments or idea semantics.

Latency prediction uses ten repeated five-fold participant splits stratified by the six design cells and inner three-fold tuning. Event counts receive a fixed log(1+x) transform before

training-only scaling. The diagnostic rating model excludes the target item. Metrics are calculated on all out-of-fold predictions in each repeat and then averaged, not averaged over small fold-specific R^2 values. We report MAE, RMSE and R^2 on each original response scale. They are not pooled across datasets. Primary bootstrap differences condition on fitted predictions; the separate voice sensitivity described above additionally resamples and refits training data.

5.3 Assignment effects and selection

Voice role contrasts use linear regression with HC3 standard errors, controlling for study and style when they vary. We distinguish all released participants, first/only-check passers, and passers of every role check. Selected comparisons describe a different sample and cannot establish the effect among latent “true perceivers.” Post-treatment filtering can break randomized comparability. [33] Paired bootstraps resample within original study × role × style cells and recompute the full and selected estimates.

Four exploratory role interactions with user attributes receive Holm correction. The latency analysis estimates original-scale means in a saturated 3×2 model with HC3 t/F inference. Its families are seven outcomes × three omnibus tests (21) and seven outcomes × four contrasts (28), each Holm-adjusted. Pointwise confidence intervals are labeled separately. These are different estimands and procedures from the source's aligned-rank-transform analysis; significance disagreements do not constitute a failed exact replication. All new inference is exploratory.

5.4 Reproducibility checks

The scripts expose input paths, seeds, sample rules, feature blocks, tuning grids and per-observation predictions. Checks cover raw hashes, unique observation keys, item range and composite agreement, note-count reconciliation, participant aggregation, complete prediction coverage, independent recalculation of prediction metrics, and rank-deficient regression designs. Independent agent reviews inspected coding, leakage risks, statistical interpretation and source provenance. Review found and repaired a constant predictor and incompatible information-criterion conventions. Such review is a quality-control step, not external peer review.

6. Results

6.1 Related judgments are strongly associated, without a clean factor solution

In the pooled voice data, intelligence alpha is 0.882 and credibility alpha is 0.891. Composite Pearson correlation is 0.744; HTMT is 0.847 with a percentile interval [0.798, 0.891]. Separate-study HTMT intervals are wider and cross 0.90. The two-factor model fits better than the one-factor model, but pooled RMSEA remains 0.105 and the latent correlation is 0.824. Thus the observations support related, potentially distinguishable judgments with unresolved residual structure. They do not establish orthogonal drivers or a well-validated two-factor measurement model.

Table 2. Continuous-item CFA diagnostics. Better relative fit does not remove substantial absolute misfit.

Sample/model	n	χ^2 / df	CFI	RMSEA	Factor correlation
E1 / one factor	353	349.0 / 35	0.873	0.160	—
E1 / two factor	353	167.5 / 34	0.946	0.106	0.820
E2 / one factor	200	199.1 / 35	0.872	0.154	—
E2 / two factor	200	118.7 / 34	0.934	0.112	0.832
pooled / one factor	553	503.9 / 35	0.875	0.156	—
pooled / two factor	553	242.5 / 34	0.944	0.105	0.824

In text co-creation, intelligence alpha is 0.914. HTMT is 0.771 [0.626, 0.891] with capable trust, 0.775 [0.640, 0.881] with reliable trust, and 0.894 [0.830, 0.957] with the adapted likeability block. Rank-based diagnostics give the same broad picture of substantial overlap. A three-factor intelligence/capable/reliable model improves CFI to 0.947, but RMSEA is still 0.106. We do not claim that these factors are independent or that the adapted instrument has been fully validated.

The seven text-latency quality ratings show a large first Pearson component (eigenvalue 4.389, 62.7% of standardized variance) and alpha 0.893. A permutation parallel analysis retains one component. Ceiling-heavy ordinal responses and shared measurement can contribute; this does not turn thoughtfulness, usefulness and trustworthiness into a validated intelligence factor.

6.2 Voice prediction transfers weakly from design and user information

The forward design-plus-user model reaches R^2=0.022, compared with 0.518 for credibility alone. Its MAE improvement over the training-mean baseline is only 0.017 points, although its conditional interval excludes zero. The user-only ablation performs somewhat better (R^2=0.055), showing that adding experimental design does not guarantee improvement after tuning. In reverse assertive-only transfer, design-plus-user R^2=0.026 and the MAE improvement interval crosses zero. No strong engineering predictor emerges from these limited variables.

Table 3. External prediction on direct intelligence ratings (1–7). Negative ΔMAE favors the model; intervals condition on fitted predictors.

Transfer/model	n test	MAE	R^2	ΔMAE vs mean	95% CI for Δ
E1 → E2 / mean	200	0.965	-0.007	0.000	[0.000, 0.000]
E1 → E2 / user	200	0.929	0.055	-0.036	[-0.059, -0.012]
E1 → E2 / design user	200	0.948	0.022	-0.017	[-0.028, -0.006]
E1 → E2 / credibility only	200	0.625	0.518	-0.341	[-0.442, -0.235]
E2 → E1 assertive / mean	180	1.037	-0.003	0.000	[0.000, 0.000]
E2 → E1 assertive / user	180	1.019	0.032	-0.018	[-0.062, 0.028]
E2 → E1 assertive / design user	180	1.025	0.026	-0.012	[-0.056, 0.033]
E2 → E1 assertive / credibility only	180	0.634	0.579	-0.402	[-0.504, -0.304]

Credibility alone gives MAE 0.625 in forward transfer and 0.634 in reverse transfer. This is a reproducible predictive association with another judgment from the same respondent and questionnaire. It is not evidence that credibility causes intelligence perceptions, that the system can predict these scores without questionnaires, or that a live voice assistant would retain this performance.

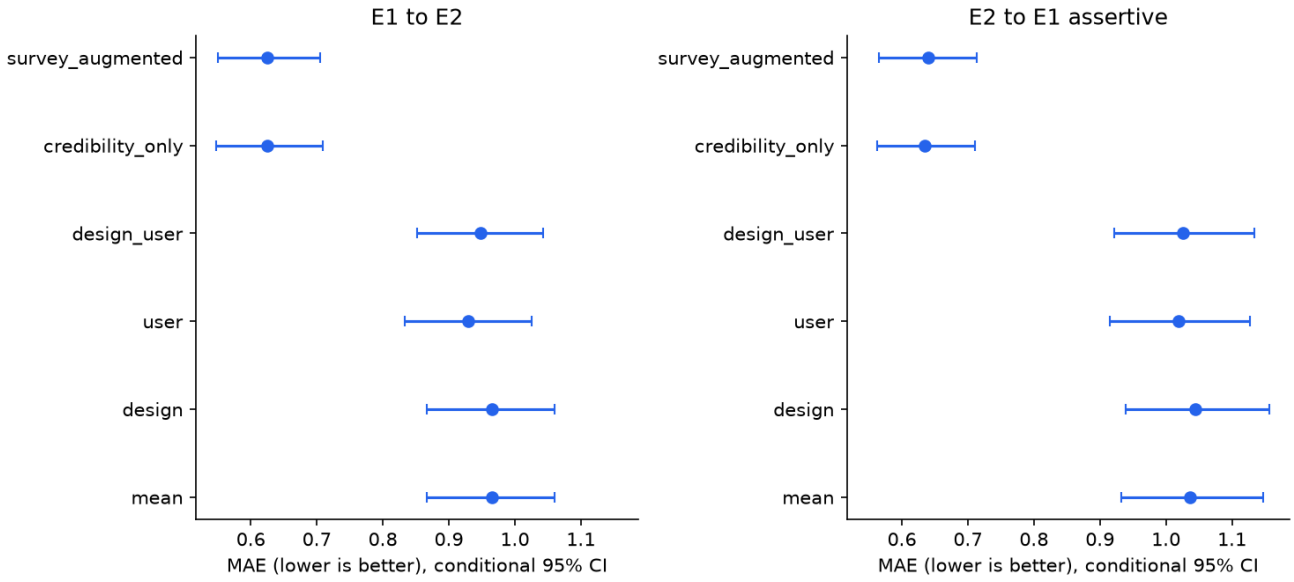


Figure 1. Study-held-out voice prediction. Concurrent credibility is a diagnostic benchmark and is unavailable in a log-only setting.

6.3 Available co-creation logs show limited point-prediction gains

For the 142 known-assignment sessions, ridge prediction using design, demographics and available logs has participant-CV $R^2=-0.005$; adding adoption gives 0.012. The nonlinear counterpart with adoption gives 0.023. Exploratory resampling ranges overlap zero, but their CV-dependent errors do not establish calibrated confidence coverage. The same-survey trust model gives $R^2=0.503$ with participant holdout and 0.512 when a condition is held out. These results survive a more flexible alternative without establishing that unmeasured behavior, semantic content or higher-quality telemetry cannot predict perceptions.

Table 4. Co-creation prediction on the 1–5 adapted intelligence scale. All models and both validation schemes are shown.

Model	Participant R^2	Condition-held-out R^2	Participant RMSE
training mean	-0.007	-0.029	0.788
design user	-0.010	0.006	0.790
telemetry	-0.005	-0.007	0.788
telemetry adoption	0.012	0.006	0.781
nonlinear telemetry	-0.117	-0.107	0.831
nonlinear telemetry adoption	0.023	-0.020	0.777
same survey benchmark	0.503	0.512	0.554
nonlinear same survey	0.382	0.394	0.618

Table 4b. E4 uses design/user as its comparator. These are paired point differences, not inferential confidence claims.

Validation	Added to design/user	ΔR²	MAE reduction
participant	telemetry	0.0047	0.0057
participant	telemetry adoption	0.0219	0.0280
condition	telemetry	-0.0131	-0.0024
condition	telemetry adoption	0.0002	0.0056

The added-log point gains are small and change with the validation scheme. Comparisons with the training mean are separate sanity baselines and cannot supply intervals for E4. Formal uncertainty for the design/user increment remains unestablished.

The association between adoption and intelligence illustrates why in-sample results should not substitute for validation. In an adjusted exploratory regression, adoption fraction has coefficient 0.802 [0.057, 1.547], but its within-model Holm-adjusted p=0.279. The corresponding predictive block gives little held-out gain. This difference reflects the distinct questions asked by coefficient estimation and generalization, along with substantial uncertainty in a small sample. Adoption can also be a consequence of the same evaluative process. We do not treat it as objective task accuracy.

A post-hoc lexical check concatenates only AI-authored ideas and adds character TF-IDF (3–5 grams, up to 10,000 features) within the training pipeline. Against a refitted design/user/log baseline, its paired R^2 increment is 0.000140 in participant CV and 0.000541 in condition holdout. The penalty grid includes 1000 and the solver is LSQR for both compared models; this reference is not numerically identical to Table 4. These negligible point increments concern one lexical representation. They do not establish that semantic quality is irrelevant, and no inferential interval is asserted.

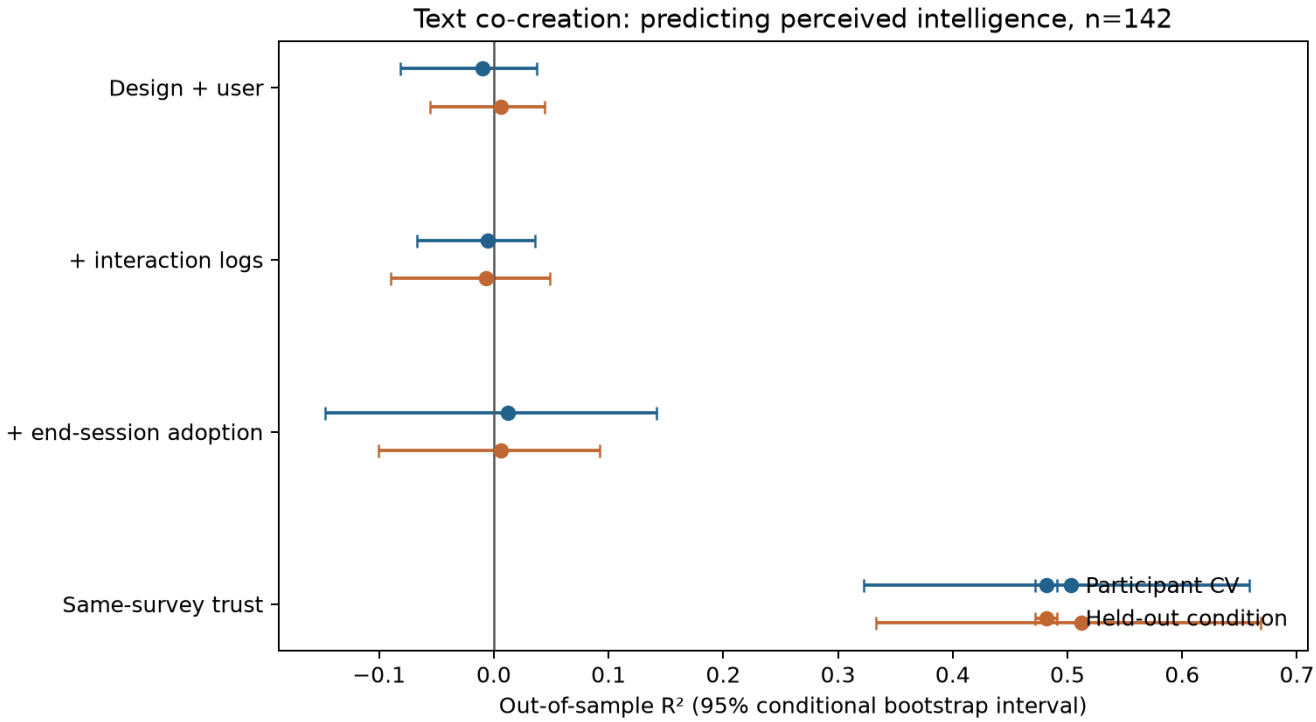


Figure 2. Ridge prediction in text co-creation. Intervals are 2,000 participant resamples of fixed out-of-fold predictions; they do not include full retraining uncertainty.

6.4 Selection changes the apparent role effect

The adjusted pooled partner-minus-servant contrast is −0.055 [−0.270, 0.159] in the 553 released eligible observations. Under the first/only-check rule, 328 participants remain and the contrast is +0.410 [0.141, 0.678]. The paired bootstrap change is +0.465 [0.275, 0.646]. This quantifies selection sensitivity, not bias relative to a known causal truth. In experiment 1, check retention is 46.4% in the partner arm versus 60.9% in the servant arm; experiment-2 first-check retention is about 69.5% in both. Equal retention rates would not alone eliminate post-treatment selection concerns.

We also reconstruct the published pooled rule: all role checks passed, combining 189 respondents from experiment 1 with 80 from experiment 2 (n=269). The observed group means exactly reproduce the source's published intelligence means, 4.9714 for partner and 4.4467 for servant. The adjusted contrast in this subset is +0.526 [0.223, 0.829]. The original authors already reported both full and selected analyses. Our contribution is an explicit reconstruction, paired sensitivity quantification and integration with measurement and prediction, not the discovery of an omitted full-sample result.

Table 5. Full and selected voice contrasts. Confidence intervals are pointwise HC3 intervals, not proof of causal subgroup effects.

Sample	Selection	n	Partner − servant	95% CI
E1	all-released	353	-0.153	[-0.428, 0.122]
E2	all-released	200	0.117	[-0.222, 0.457]
pooled	all-released	553	-0.055	[-0.270, 0.159]
pooled	first/only-MC-pass	328	0.410	[0.141, 0.678]
pooled	all-MC-pass	269	0.526	[0.223, 0.829]

A cumulative-logit sensitivity on the direct intelligent item retains the distinction between full and selected analyses. It is a one-item check under proportional odds, not ordinal validation of the full scale. None of four exploratory role-by-user interactions survives Holm adjustment. This does not establish equality of user groups or exclude smaller moderators; the ownership-frequency analysis is restricted to respondents with relevant service-use answers.

6.5 Latency effects depend on the endpoint and inferential specification

In the 240-person text study, Creation receives usefulness ratings 0.353 points higher than Advice, averaged equally across assigned delays (95% CI [0.146, 0.561]). This survives both stated Holm families. Average longer-delay thoughtfulness is 0.336 points higher than at two seconds, with a pointwise interval [0.037, 0.635], but Holm p=0.718 over 28 contrasts. No delay-by-task interaction is rejected after correction. These results do not establish a zero latency effect and cannot recommend delaying voice responses.

Table 6. Original-scale latency-study contrasts. Holm adjustment covers 28 planned exploratory contrasts.

Outcome/contrast	Difference	95% pointwise CI	Raw p	Holm p
thoughtfulness / 9 minus 2	0.323	[-0.010, 0.656]	0.057	1.000
thoughtfulness / 20 minus 2	0.349	[0.026, 0.672]	0.034	0.859
thoughtfulness / long minus 2	0.336	[0.037, 0.635]	0.028	0.718
usefulness / long minus 2	0.249	[-0.001, 0.499]	0.051	1.000
usefulness / creation minus advice	0.353	[0.146, 0.561]	0.001	0.026

Excluding the three ambiguous task IDs changes these key contrasts little. With 237 participants in prediction, telemetry R^2 is -0.006 for thoughtfulness and -0.017 for usefulness. Other concurrent survey ratings give 0.500 and 0.547 respectively. The replicated information-source pattern is suggestive but remains triangulation across different constructs. It is not external validation of an intelligence model on the latency dataset.

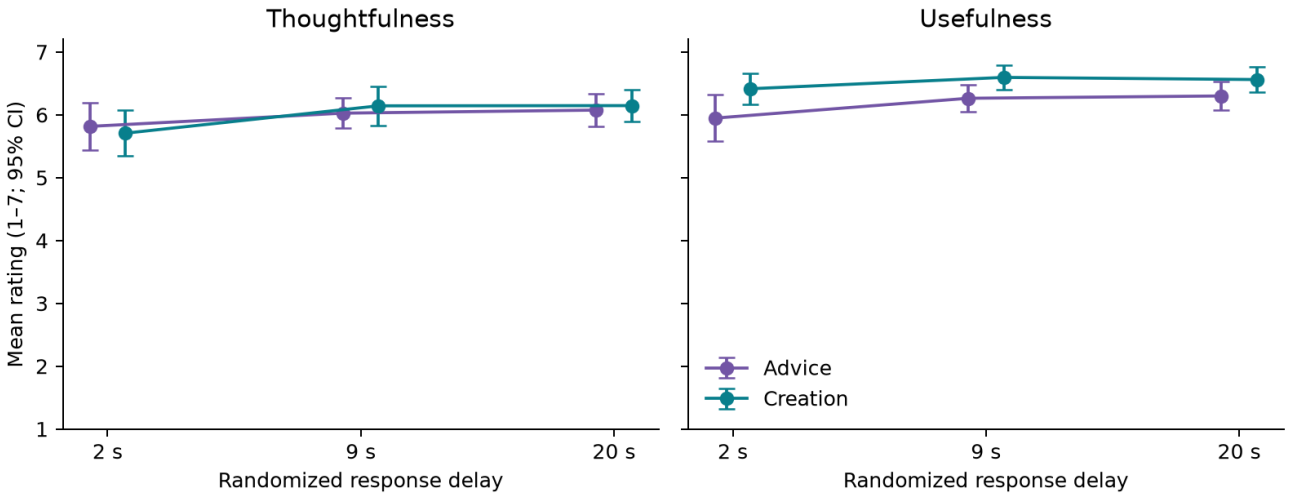


Figure 3. Text-study cell means and pointwise 95% intervals. The complete 1–7 response scale is shown; this is not a voice experiment.

6.6 Paired contrasts quantify the information-source gap

A post-hoc analysis compares predictions for the same held-out people, using 4,000 paired bootstrap resamples. It contrasts the design-plus-user model against credibility in the voice studies, design/user/logs against trust in co-creation, and design/logs against other quality ratings in text chat. R^2 gain is the reduction in squared error divided by the same test-set outcome variance. Reported user traits are themselves survey data; the distinction is absence versus presence of concurrent evaluative judgments, not an absolute objective/subjective divide.

Table 7. Paired gain from concurrent evaluations over the stated comparison feature block. No pooled cross-study estimate is calculated.

Validation/endpoint	n	R^2 gain	2.5–97.5% resampling range
Voice E1_to_E2	200	0.496	[0.335, 0.631]
Voice E2_to_E1_assertive	180	0.553	[0.426, 0.669]
Co-creation participant	142	0.508	[0.331, 0.670]
Co-creation condition	142	0.519	[0.330, 0.698]
Thoughtfulness (text)	237	0.503	[0.375, 0.620]
Usefulness (text)	237	0.545	[0.391, 0.659]

All six fixed-prediction resampling ranges are above zero. For text CV, overlapping training sets make OOF errors dependent: these ranges are exploratory reweighting summaries, not calibrated 95% confidence intervals or six significance tests. Their coverage has not been established. The independent voice holdouts and separate training bootstrap provide a different uncertainty design. The observations and validation schemes overlap within datasets; these are not six independent replications. Latency contrasts use each participant's predictions averaged over CV repeats, so their R^2 contrast differs slightly from subtracting average per-repeat metrics. The result does not identify why concurrent ratings are informative and does not prove that richer telemetry would fail.

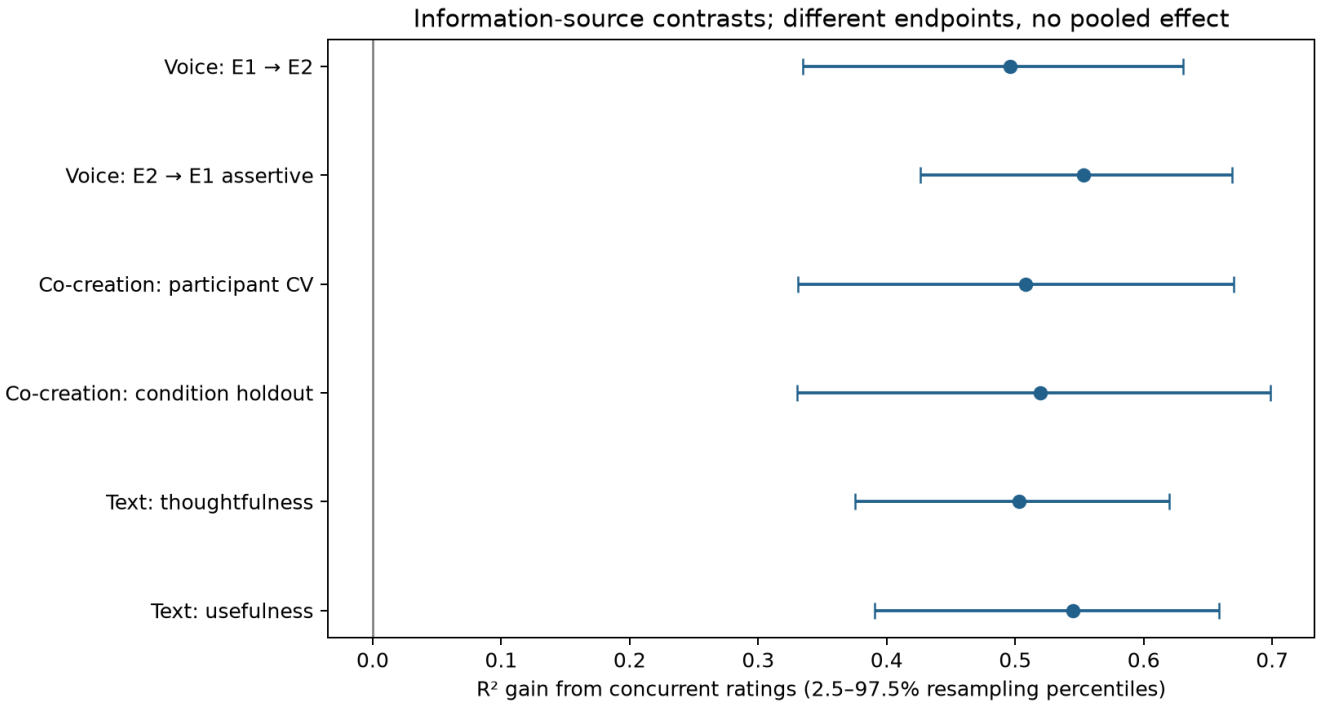


Figure 4. Paired information-source comparisons. Each row retains its own outcome and validation unit. Ranges hold predictions fixed. Text CV error dependence is not accounted for; coverage is unestablished.

6.7 The main gap persists when models are retrained

The training-and-test bootstrap preserves the advantage of concurrent credibility in both voice transfers. In contrast, the small design-plus-user MAE advantage over the training mean has an interval spanning zero in both directions. The forward lower endpoint is only −0.0002: this borderline crossing should not be treated as a decisive significance reversal, especially with Monte Carlo uncertainty. The evidence supports a large information-source contrast and a small, uncertain operational improvement. The narrower conditional interval in Table 3 does not represent uncertainty over the whole training procedure.

Table 8. One thousand paired training-and-test bootstrap draws per transfer. Positive values favor the first named information set.

Transfer / contrast	Grouped-CV reference	95% train-and-test interval
E1 to E2 / MAE reduction: credibility vs design/user	0.324	[0.2108, 0.4285]
E1 to E2 / R^2 gain: credibility vs design/user	0.496	[0.3205, 0.6314]
E1 to E2 / MAE reduction: design/user vs mean	0.017	[-0.0002, 0.0667]
E2 to E1 assertive / MAE reduction: credibility vs design/user	0.389	[0.2820, 0.4908]
E2 to E1 assertive / R^2 gain: credibility vs design/user	0.551	[0.4229, 0.6853]
E2 to E1 assertive / MAE reduction: design/user vs mean	0.012	[-0.0411, 0.0577]

These intervals use empirical resampling within the same two studies. They are not population-wide guarantees, and the two directions reuse the source datasets. Reference estimates can differ slightly from Table 3 because grouped rather than shuffled inner folds are needed to keep duplicated bootstrap participants together. The observed robustness supports the main predictive contrast under a stronger uncertainty analysis, while leaving causation and live-voice generalization unresolved.

7. Discussion

7.1 What has been learned about perceived “smartness”

Across the three source releases, predictions from experimental conditions or coarse logs differ substantially from predictions using nearby subjective judgments. Speed, correctness and politeness are not jointly observed, so their relative importance cannot be established here. The latter captures a large share of variance, while the former provides small, unstable or absent gains under the tested models. The resulting operational lesson is to state the information budget whenever reporting a perception predictor.

This pattern has several possible explanations. The available logs omit whether an answer was actually right, contextually apt or successfully repaired. Participants may combine several cues into a common evaluation. Items such as competent, knowledgeable and trustworthy overlap in meaning. A shared response style can also correlate judgments. Our designs cannot separate these explanations. Calling the gap “common-method bias” as an established cause would overstate the evidence; calling credibility an objective driver would make the opposite error.

The measurement results sharpen the interpretation. High reliability coexists with substantial between-scale association and residual factor-model misfit. Multiple dimensions may be useful without being independent, and a high alpha does not determine how many mechanisms exist. Researchers should report absolute fit, residual limitations and cross-group equivalence before interpreting a latent score as the same quantity across scenarios. The data do not support fixed universal factor weights for contemporary spoken assistants.

7.2 Context, user differences and causal interpretation

Context enters both the theoretical specification and validation. Holding out a whole voice experiment or co-creation condition tests a different generalization claim from random participant folds. In these releases, strong contemporaneous survey association transfers more readily than design-based prediction. This does not establish that the psychological construct itself is invariant. The voice experiments share a laboratory, country and stimulus family, while the co-creation conditions share a task and system.

User-only information has limited forward predictive value, and the tested interactions are uncertain. That is insufficient to conclude that user characteristics do not matter. More precise prior-experience measures, accessibility needs, language proficiency and repeated-use history may be important, but this study cannot estimate them. The fitted demographic coefficients are descriptive; they are not a recommendation to personalize treatment based on protected categories.

Selection deserves equal attention to model fit. A model of perceived intelligence often contains recognition checks or user interpretations that occur after exposure. Excluding participants based on those measures can alter the role contrast markedly. Full-release and selected analyses answer different questions. Reproducing a selected-sample effect is valuable, but it does not recover a population assignment effect or identify how much of a treatment's influence passes through recognition.

7.3 Implications for evaluation practice

For a product evaluation, collect direct intelligence judgments separately from satisfaction and trust. Preserve objective success/correctness and timing even when users report a good experience. Declare which inputs are available before, during and after a session. Compare against a training-mean baseline and a simple established model before introducing a complex architecture. Use participant, scenario and system holdouts, with all tuning confined to training. Treat same-survey predictors as a separate diagnostic analysis.

These recommendations do not imply that perception should be maximized regardless of actual capability. A system can become more persuasive without becoming more correct. The present evidence supports tracking perceived competence and verified performance as separate outcomes; their alignment requires new data. Neither the null telemetry result nor an attractive subjective score justifies suppressing errors, adding artificial delays or replacing correctness checks with a perception metric.

8. Limitations and unperformed tests

Modality and ecological validity. The only direct voice-specific outcome comes from observers of video vignettes. There are no live speech-recognition errors, interruptions or per-turn repair measurements. The actual interaction datasets are text/canvas and text chat. Claims about real-time spoken interaction remain prospective.

Measurement. Instruments differ in item wording, response range and administration. Continuous covariance models have poor absolute fit. Full ordinal CFA, measurement invariance, test–retest stability, Russian translation validation and a distinction between general halo and causal mechanisms were not completed. The latent specification is a falsifiable proposal; its complete parameters were not estimated.

Available information. No analyzed release jointly measures correctness, task success, true response latency, repair quality, user expectations and intelligence. Coarse event counts and surviving-note statistics may be weak proxies. Language-model semantic encoders and large neural predictors were not evaluated; the negative result is limited to the specified features, ridge models and the co-creation nonlinear sensitivity.

Sampling and inference. Public releases already include exclusions. Six co-creation assignments are unknown and three latency IDs are ambiguous. The source studies are not a probability sample of assistant users. Primary conditional intervals omit model-training uncertainty; the voice retraining sensitivity addresses empirical training variation but not study or stimulus sampling. Co-creation and latency were not subjected to full retraining bootstraps; row-resampling of their fixed OOF predictions does not establish confidence coverage for dependent CV errors. Exploratory findings require independent confirmation. No prospective sample-size calculation can retroactively make this secondary analysis confirmatory.

Related work. The literature search is broad but not exhaustive, and several full texts were unavailable. The methods are established; the contribution is their application to these datasets. No external peer review, venue acceptance or blanket “first-ever” claim is made. A strong submission would benefit from prospectively collected live-voice evidence and human domain-expert review.

9. Competing mechanisms and prospective discrimination

Status: proposals only. The expanded primary-source review of 5 November 2026 does not establish worldwide priority for any candidate. We no longer treat clarification as the predetermined main contribution. The original question is why interaction creates an attribution of intelligence beyond the final task outcome, and whether this attribution is distinct from usefulness or general favorable evaluation. The new review compares memory, correction, initiative, expectations and allocation of effort; language, modality and model count alone are insufficient novelty.

9.1 Close prior work substantially narrows the claim

Memory references already affect direct intelligence ratings; recovery style has been studied with successful correction, and selective correction versus collateral changes has prior robot-learning evidence. These are serious predecessors, not adjacent background that can be dismissed by changing the modality. [34] [35] [36]

Human generalization across LLM tasks, repeated belief updates and delegation already have explicit studies. Dreyfuss and Raux additionally compare similarly unhelpful chatbot failures that differ in perceived reasonableness. Pohl and colleagues demonstrate that minimal behavioral cues can influence multiple attributed traits. These sources make global halo, task similarity and interpretation of evidence strong competing explanations. [37] [38] [39] [40]

Mixed initiative is longstanding, and modern proactive agents are already evaluated for intelligence and timing. Redundant questions and human dissatisfaction also have direct precedents. A recent October 2026 preprint reports a human clarification-efficiency study; it does not measure the same perceived-intelligence endpoint. The companion review records full-text sections, indexed-text versus abstract access, publication dates and remaining access-limited threats. [41] [42] [43] [44] [45]

9.2 What Zheng and colleagues do and do not isolate

Zheng's text experiment elicits unspecified preferences while independently supplied hard constraints can be violated. Questions also change personalization; unasked preferences are randomly selected. Section 5.2 explicitly leaves grounding and personalization unresolved. Its measures include alignment, trust and satisfaction rather than direct intelligence. Replacing the outcome label would not create a substantial contribution. [46]

9.3 Corrected clarification contrast

When a parameter is available, the baseline acknowledges and uses it. When absent, the baseline states a prespecified default as an assumption. A request either asks again or obtains missing information; a confirmation either displays the known value or proposes the default for approval. Thus the policy labels conceal different state-dependent actions unless these are made explicit. Broad repetition of the whole request is an additional effort manipulation and is removed from the core contrast.

The known-state request-minus-use and missing-state request-minus-assumption contrasts are valid conditional assignment effects. Their difference is heterogeneity of a contingent policy effect, not an isolated grounding mechanism. Success and effort may be consequences. Selecting successful sessions or adjusting for post-treatment usefulness cannot manufacture an equal-outcome effect. A separate known-information stimulus experiment can randomize acknowledgment/use, explicit confirmation and bare re-request while fixing the final answer by design. Confirmation-versus-request is narrower, but wording, pragmatic purpose and burden still require checks.

9.4 Proposed experiment: work allocation and future expectations

The implemented text pilot presents two correctly completed operations: updating dependent quantities and selecting an option under a constraint. The assistant performs one operation and a separate module performs the other. Assignment is randomized while the total amount of delegation, arithmetic information and final answer remain fixed. Future access to the module is randomized independently. Participants forecast success on four tasks crossing topic and operation, and separately rate perceived intelligence, liking, usefulness, reading difficulty and the assistant's credit for the result. Credit is measured after forecasts to avoid introducing the authorship question beforehand. These are prepared fictional scenarios, not observed traces from deployed models.

The main contrast compares how swapping the executors changes forecasts for each operation. Both absolute operation contrasts and their difference are retained. However, a positive difference, and even opposite signs for the two operations, do not uniquely identify operation-specific capability inference. The accompanying mathematical examples produce this pattern from a mixture of opposite global shifts across latent respondent types. Taking logits of group means does not remove that ambiguity. The examples are not human observations.

Attribution of authorship and capability remain coupled in this manipulation. A further experiment would need to hold authorship fixed while independently supplying evidence of unaided performance. Before confirmatory collection, external participants must establish whether the scenarios are understood, whether task difficulty leaves room for changes in forecasts, and whether the intelligence measure is distinct from neighboring judgments. Competing models and a meaningful effect or precision target must be specified before choosing sample size. If forecasts change without a direct intelligence effect, the result concerns expectations about performance rather than an explanation of perceived intelligence. No human results from this pilot are reported here.

9.5 Observation, coding and feasibility boundaries

The codebook stores an episode's ordered turns, candidate question/parameter units and the observed input before each assistant question. User-provided information is separate from assistant-available information. Questions have explicit new-information, repeated-known, confirmation, other or unknown labels and bounded evidence references. Without an observed input snapshot, availability remains unknown; ASR loss cannot be silently attributed to reasoning failure. The software checks declared provenance and chronology, not semantic truth or actual log authenticity.

Author diaries are development material. Authors can remember their own episodes even if brand and rating metadata are removed. Packets therefore record coder familiarity, and reports separate development versus held-out material, author versus external participants, modality and provenance. After freezing the codebook, new external episodes and independently recorded labels are needed for validation. Agreement on fixed candidate units does not prove exhaustive question detection, construct validity or population reliability. Legacy records retain an explicit lack-of-detail status rather than invented turns.

With zero monetary budget, locally scripted text, recorded audio or an accurately described Wizard-of-Oz feasibility pilot is possible if external volunteers are actually available. No such recruitment, new human observations, translated-scale validation or confirmatory power calculation has occurred. Fix the smallest meaningful effect and precision target, then simulate the chosen clustered design before main collection. Eight browser conversations about timers were used to check access and prompts, not to rank models or measure human perception. They used ChatGPT, DeepSeek, Qwen and search-enabled Perplexity; access and settings must be rechecked for a new collection. Full live-voice transfer remains future work.

10. Conclusion

We complete a reproducible secondary empirical study of directly measured perceived intelligence and related outcomes, with explicit measurement, prediction and selection checks. Design and limited behavioral logs provide much less predictive information than contemporaneous subjective judgments in these releases. Correlated-factor models improve relative fit but do not validate independent components, and role effects depend materially on post-treatment selection. The evidence supports a model whose interpretation depends on context and information availability. It does not support a universal engineering score for live voice-assistant “smartness.” The next decisive test requires the measurements that the available datasets do not jointly contain.

Transparency and artifacts

The repository includes raw-source manifests and downloader, pinned Python dependencies, analysis programs, selected aggregate outputs, an inference example, literature records and the prospective pilot. Individual predictions and full derived outputs are regenerated locally and excluded from version control. The research plan, codebook, observation tools and comparison planner support future collection. Automated checks include synthetic SPSS-to-CSV tests. AI assistance was used for literature retrieval, programming, analysis and drafting. No new participants were recruited for these tools. Raw-data rights remain with their source authors.

Analysis date: 4 November 2026. Primary seed: 20261104; training-bootstrap seed: 202611041. Reproducibility entry point: python scripts/run_all.py. The article tables are generated from saved results, while all conclusions are checked against those same outputs. Detailed sensitivity tables remain available even when not printed in the main article.

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